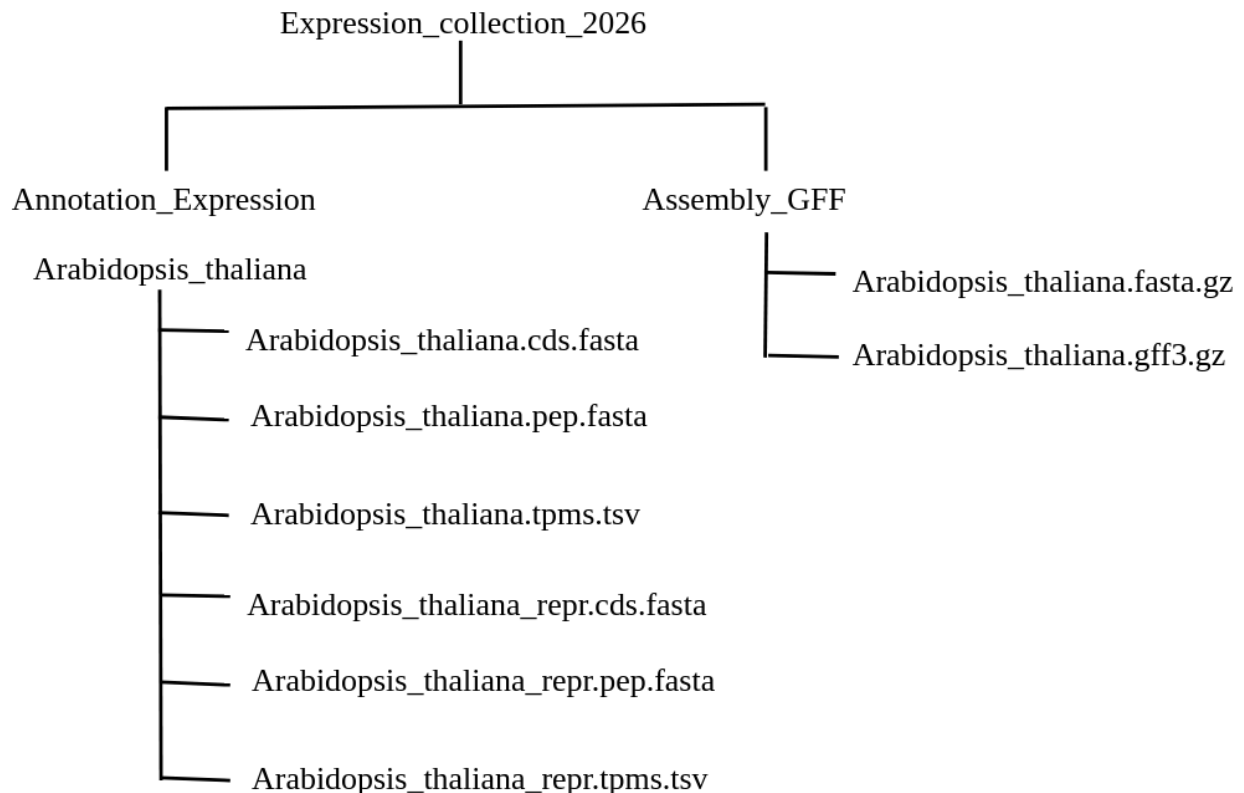


Guidelines for collecting and storing gene expression data

- Kindly use the following filters for selecting the SRA files from NCBI:
 - Click on the species name in the taxonomy box on the right of the page to restrict only the specific species samples
 - Source: RNA
 - Library: Paired
 - Platform: Illumina, BGISEq
 - File type: fastq
- Please make sure to choose the best available annotation for the species when you collect its SRA files for making the expression file; Please also make sure that they are not restricted in any way.
- Generate the PEP files from CDS to ensure that the gene IDs are matching
- Assess the BUSCO completeness score of the PEP file using the BUSCO protein mode with embryophyta_odb12 lineage for the flowering plants - Make sure to do the BUSCO analysis on the PEP file with and without multiple isoforms; If you are processing multiple species, best way is to use the BUSCO batch mode that processes efficiently.
- I have created a private GitHub repo to organize the scripts that can be of use in the gene expression collection initiative: https://github.com/PuckerLab/Xpression_collect Some of these are my own scripts and some are from Boas. If you already have scripts for these steps, please feel free to use them. I just wanted to have them in one place to make it more organized.
- If you are on GitHub already, I have added you to the repo as admins. Please let me know if you cannot access it. If you don't have a GitHub repo yet, kindly create one and share your username with me, so that I can add you as well.
- Make sure to obtain the expression TPM and count files using CDS with isoforms; Use the isoform_purge_wrapper.py script (in the Xpression_collect repo) to obtain CDS, PEP, expression file with only primary transcripts.
- Please make sure to regularly update the tracker sheet for the gene expression dataset collection regularly - [Xpression_collect_tracker](#)
- A few pointers on the tracker - kindly provide the database link to access the specific annotation files easily (Database access link) apart from the reference citation; please update the date on which you retrieved the SRA files for the particular species (Date of SRA data retrieval) which will act as the time stamp for updating the data publication later; Importantly fill the status column for species that you are currently processing, so that there is no duplication of efforts and we can make the collection process more efficient, and well coordinated
- Some pointers on file names - Let us have the extension .cds.fasta, .pep.fasta, .fasta, .tpms.tsv, .counts.tsv to respectively denote coding sequence, peptide sequence, assembly, TPM, count expression files
- General workflow to obtain expression file per species - Fetch the relevant high quality annotation (CDS, PEP) for the species, and SRA files from NCBI; Run the wrapper script for SRA dataset fetching, kallisto run, kallisto output merge; Then use the filter RNASeq

script to remove genomic and low quality samples from the expression TPM file - note that this needs the corresponding counts file as well; Run the isoform purge wrapper script on the expression file, and CDS file to obtain CDS, PEP, expression file with only primary transcripts. The workflow representation and model commands can be found in the GitHub repo.

- One VM from my PlantExpCollect project will be used as a storage hub for the expression data.
- The folder architecture for storage in this VM will be as follows:



The _repr suffix denotes files with only primary transcripts

- Kindly follow the naming conventions and folder organization strictly, so that the collection is uniform and it will be easier for the data publication
- Please note that the Assembly_GFF folder is for our internal research purposes and won't be a part of the publication. We thought it would be easier if we could access the corresponding assembly and GFF files easily for downstream analyses like synteny and gene duplication finding that need positional information. Also please compress the assembly and GFF files to save space
- Save the expression files as tsv files as shown in the sample folder organization